

Car Wash

Carpets

Sofas

Cleanly

Platform Blueprint

Complete Build Specification — Web + Mobile
Arabic & English · UAE · Saudi Arabia · Egypt

VERSION

v1.0 — Ready to Build

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SERVICES

Car Wash · Carpets · Sofas

MARKETS

UAE · Saudi Arabia · Egypt

LANGUAGES

Arabic (RTL) + English

PLATFORMS

Web + iOS + Android

TEAM

4–6 people · 6 months

1 PRODUCT OVERVIEW

Cleanly

On-demand mobile cleaning services — the washer always comes to the client.

Car Wash	Carpets	Sofas & Furniture
Exterior · Interior · Full Detail Per vehicle · Add-ons: tire shine, wax	Standard · Steam · Stain treatment Per room · Add-ons: deodorizer, protection	Deep clean · Stain removal · Polish Per seat/piece · Add-ons: fabric protection

Key Product Decisions

Decision	Choice	Reason
Business model	Marketplace — client picks company	Gulf customers trust brand names; faster to launch than gig-dispatch
Who moves?	Washer travels to client	Maximum convenience — car/home/office service anywhere
Languages	Arabic (RTL) + English	UAE, KSA, Egypt primary markets — dual-language from day 1
City filtering	Company sets coverage cities	Client GPS → matched to companies serving that city
Platforms	Web + iOS + Android	Customer: web + mobile · Company: web · Washer: mobile only

2 ARCHITECTURE & TECH STACK

5 App Surfaces

Surface	Framework	Deploy	Users
Customer web	Next.js 14 (App Router) + Tailwind	Vercel	End customers — booking + tracking
Customer mobile	React Native + Expo Router v3	App Store + Play Store	End customers — iOS & Android
Company web	React + Vite SPA + TanStack Query	Vercel	Company admins — ops + analytics
Washer mobile	React Native + Expo (shared codebase)	App Store + Play Store	Field washers — GPS + camera + checklist
Admin web	Next.js + NextAuth (Google SSO)	Vercel (separate domain)	Platform team — verification + control

Backend Stack

Layer	Tool	Why chosen over alternatives
API server	Node.js + Fastify	2x faster than Express · built-in Zod schema validation · plugin architecture
ORM	Prisma	Type-safe queries · migrations as code · Neon adapter built-in
Primary DB	Neon PostgreSQL + PostGIS Serverless	auto-scale to zero · DB branching for staging · Bahrain region
Cache / sessions	Upstash Redis (serverless)	Pay-per-request · rate limiting · washer location cache (5s TTL)
Async jobs	BullMQ + Upstash	Payout processing · email queues · photo processing · retry logic
Real-time	Socket.io	Washer GPS → customer map · order status push · rooms per order
File storage	Cloudflare R2	Zero egress fees · CDN delivery · S3-compatible · before/after photos
Payments	Stripe + Stripe Connect	AED currency · Apple/Google Pay · company payouts · no PCI scope
Push notifs	Expo Push (FCM + APNs)	Single API for iOS + Android · token management handled
SMS	Twilio Verify + SMS	OTP auth · order confirmations · Gulf number coverage excellent
Email	Resend + React Email	Transactional · bilingual templates · developer-friendly
WhatsApp	360dialog (Phase 2)	Critical for Gulf market · high WhatsApp usage rate
Monitoring	Sentry (all 5 surfaces)	Error tracking · performance tracing · release tracking
CI/CD	GitHub Actions + Turborepo	Lint+test+typecheck per PR · auto-deploy on merge · monorepo aware

Monorepo Structure (Turborepo)

```
/cleanly-monorepo ← Turborepo root
apps/
customer-web/ ← Next.js 14 — booking + tracking
customer-mobile/ ← Expo — iOS + Android
company-web/ ← Vite React — company dashboard
washer-mobile/ ← Expo — GPS + camera + checklist
admin-web/ ← Next.js — admin panel
api/ ← Fastify — REST + Socket.io
packages/
```

shared-types/ ← TypeScript interfaces (all apps)
shared-utils/ ← Price calc, date format, AR/EN helpers
ui-components/ ← Shared React components
i18n/ ← All AR/EN translation strings
config/ ← ESLint, Tailwind, TS configs
prisma/ ← Single schema · migrations · seed
.github/workflows/ ← CI/CD pipelines per app

3 DATABASE SCHEMA (15 TABLES)

SERVICE_CATEGORIES

- id (PK uuid)
- slug (unique)
- name_en
- name_ar
- icon
- is_active bool

CITIES

- id (PK uuid)
- name_en
- name_ar
- country_code
- is_active bool

CUSTOMERS

- id (PK uuid)
- full_name
- phone (encrypted)
- email
- preferred_lang (en/ar)
- wallet_balance float
- loyalty_points int
- created_at

COMPANIES

- id (PK uuid)
- name_en
- name_ar
- description_en
- description_ar
- logo_url
- avg_rating float
- total_reviews int
- is_active bool
- is_verified bool
- stripe_account_id

COMPANY_CITIES

- id (PK uuid)
- company_id (FK)
- city_id (FK)
- is_active bool

COMPANY_CATEGORIES

- id (PK uuid)
- company_id (FK)
- category_id (FK)

PACKAGES

- id (PK uuid)
- company_id (FK)
- category_id (FK)
- name_en
- name_ar
- description_en
- description_ar
- unit_type (per_vehicle/per_room/per_seat)
- base_price float
- duration_minutes int
- is_active bool

ADDONS

- id (PK uuid)
- company_id (FK)
- category_id (FK)
- name_en
- name_ar
- price float
- is_active bool

WASHERS

- id (PK uuid)
- company_id (FK)
- full_name
- phone
- avg_rating float
- total_jobs int
- is_online bool
- current_lat
- current_lng
- last_seen

ORDERS

- id (PK uuid)
- customer_id (FK)
- company_id (FK)
- washer_id (FK)
- package_id (FK)
- category_id (FK)
- city_id (FK)
- status (confirmed/en_route/arrived/in_progress/completed/cancelled/disputed)
- quantity int
- service_lat
- service_lng
- location_note
- subtotal
- addons_total
- platform_fee
- tip
- total
- payment_status
- placed_at
- arrived_at
- completed_at

ORDER_ADDONS

- id (PK uuid)
- order_id (FK)
- addon_id (FK)
- price_at_time float

REVIEWS

- id (PK uuid)
- order_id (FK)
- customer_id (FK)
- company_id (FK)
- rating int (1-5)
- comment
- created_at

PAYMENTS

- id (PK uuid)
- order_id (FK)
- stripe_payment_intent
- amount float
- method (card/apple_pay/wallet)
- status
- paid_at

PAYOUTS

- id (PK uuid)
- company_id (FK)
- gross_amount
- platform_commission
- net_amount
- status (pending/processing/paid)
- period_start
- period_end
- paid_at

PROMO_CODES

- id (PK uuid)
- code (unique)
- discount_type (pct/fixed)
- discount_value float
- max_uses int
- used_count int
- category_id (FK nullable)
- city_id (FK nullable)
- expires_at

Key Indexes (defined in Prisma schema from Day 1)

Index	Table.Column(s)	Query it serves
idx_orders_customer_status	orders(customer_id, status)	Show my active orders — most frequent customer query
idx_company_cities_city_active	company_cities(city_id, is_active)	Home screen company list — runs on every app open
idx_washers_company_online	washers(company_id, is_online)	Company sees available staff for assignment
idx_orders_company_placed	orders(company_id, placed_at DESC)	Company live feed and analytics queries
idx_reviews_company	reviews(company_id)	Compute avg_rating per company
idx_packages_company_category	packages(company_id, category_id, is_active)	Package list when customer views company profile

4 ORDER LIFECYCLE (10 STEPS)

1 Customer pins location + picks service category

GPS coords saved · parking/address note captured · service category selected (car wash / carpet / sofa) · city auto-detected from GPS

2 Customer browses and selects company

API query: companies WHERE city_id matches AND category matches AND is_active AND is_verified · sorted by avg_rating · customer sees price, ETA, reviews

3 Customer selects package + add-ons

Package list filtered by category · quantity selector shown (rooms for carpets, seats for sofas) · add-ons displayed per category

4 Customer pays

Stripe PaymentIntent created · card / Apple Pay / Google Pay / wallet charged · order row inserted status=confirmed · payment row inserted · SMS + email confirmation sent in preferred language

5 Platform notifies company

FCM push to company web dashboard + Socket.io event · order appears in live feed · company has 60s to assign a washer before escalation alert

6 Company assigns washer

Company selects available washer from staff list · washer FCM alert on their app · 30s accept window · if declined → next available washer · order status → assigned

7 Washer navigates to location

Washer app opens Google Maps to service GPS pin · washer location broadcast every 5s via Socket.io → customer sees live dot on map · order status → en_route

8 Washer arrives + uploads before photo

arrived_at timestamp saved · before photo uploaded to Cloudflare R2 (timestamped + GPS-tagged) · customer push: Your washer has arrived · order status → arrived

9 Wash / clean in progress

Per-category checklist shown to washer · items toggled done · ETA countdown on customer screen · order status → in_progress

10 Job complete — payout triggered

Washer taps done · after photo uploaded to R2 · completed_at saved · order status → completed · customer push + SMS · review prompt shown · tip charged · platform fee deducted · net amount queued for weekly company payout via Stripe Connect · loyalty points credited

Order Status State Machine

Status	Trigger	Description
confirmed	→	Payment captured · company notified
en_route	→	Washer accepted · GPS broadcast live

arrived	→	Before photo uploaded · timer starts
in_progress	→	Checklist active · ETA countdown
completed	→	After photo · payout queued · review
cancelled	→	Full refund if before arrival · partial after
disputed	→	Admin reviews photos · manual decision

5 SECURITY BLUEPRINT

Authentication — Per Role

Role	Method	Token	Extra protection
Customer	Phone OTP (Twilio Verify)	JWT 15min + refresh 30d in HttpOnly cookie	Device fingerprint · new device alert · 3 OTP attempts then lock
Company admin	Email + bcrypt password	JWT + TOTP MFA (Google Authenticator)	MFA required for payout access · forced password reset on failed login
Washer	Phone OTP (Twilio)	JWT + 4-digit PIN	Company admin creates account · PIN set on first login
Platform admin	Google Workspace SSO only	NextAuth session	Hardware MFA (YubiKey) · IP allowlist at Vercel edge · full audit log

API Security Layers

Layer	Implementation	Config
Rate limiting	Upstash Redis sliding window	OTP: 3/phone/15min · Orders: 10/user/hr · General: 100/min authenticated · 20/min unauthenticated
Input validation	Zod schema on every endpoint	Type + length + pattern checked before any handler runs · 400 on schema fail · no silent pass-through
SQL injection	Prisma parameterized queries	No raw SQL from user input · \$queryRaw with tagged templates only when necessary
XSS prevention	DOMPurify + CSP headers	All rich text sanitized on write and read · Content-Security-Policy set via Helmet.js
CSRF	SameSite=Strict + Double Submit Cookies: SameSite=Strict	API: CSRF token for state-changing mutations on web
Secrets	Vercel + Railway env vars	Never in code · rotated quarterly · different keys per environment · GitLeaks in CI
HTTPS	Enforced everywhere	HSTS header · redirect all HTTP → HTTPS · min TLS 1.2

Data Protection & Compliance

Area	Requirement	Implementation
Payment data	PCI DSS scope reduction	Stripe Elements handles card fields · platform stores only last4 + brand + Stripe payment method ID
Stripe webhooks	Authenticity verification	stripe.webhooks.constructEvent() on every webhook · reject any event without valid signature
Phone numbers	Encryption at rest	AES-256 encrypted in Neon · decrypted only in application layer · never written to logs
UAE PDPL	Data protection compliance	Consent on signup · data minimization · right to deletion · privacy policy in Arabic + English
Saudi PDPL	KSA data rules	User consent for marketing comms · opt-out respected · data localization consideration
Photos	Immutable evidence	R2 signed URLs only (1hr expiry) · timestamped + GPS-tagged on upload · immutable after written
Reviews	Fraud prevention	Review only allowed after verified completed order · one review per order · admin moderation queue

Threat Model — Top Risks

Threat	Severity	Mitigation
SMS OTP billing fraud	Critical	Rate limit 3/phone/15min · Twilio fraud scores · phone blocklist
Fake orders (no-show)	Critical	Payment captured upfront · cancellation penalty policy · photo evidence required
Washer GPS spoofing	High	Plausibility check: max speed threshold 120km/h · jump detection alert to company

Account takeover	High	New device OTP re-verify · session invalidation on password change · suspicious login alert
Fake reviews	High	Review gated behind completed+verified order ID · one per order · report button
Photo manipulation	High	Photos GPS-tagged + timestamped on upload · stored immutably · admin has access for disput
Competitor scraping	Medium	Rate limiting on listing endpoints · Cloudflare bot protection · login required for pricing detail
SQL injection	Low	Prisma ORM + Zod validation — mitigated by design, no raw SQL from user input

6 SCALABILITY ARCHITECTURE

Growth Tiers

Tier	Volume	Infrastructure	Est. Cost/mo
1 — Launch	0–500 orders/day	Single Railway container · Neon free (scale-to-zero) · Upstash free · Vercel hobby · Socket.io co-located	\$50
2 — Growth	500–5,000 orders/day	Railway 2 replicas · Neon Pro + read replica · Upstash paid · R2 CDN as S3 · Socket.io separate service	\$800
3 — Scale	5,000+ orders/day	AWS ECS Fargate autoscaling · Aurora Serverless PostgreSQL · Elasticache Redis cluster · Socket.io Re	\$2,000

Real-Time Scalability (Socket.io)

Pattern	Implementation	Why it matters
Room per order	Each order gets a Socket.io room ID · washer + customer servers only join 2 people — not broadcast to all connected	GPS events only reach 2 people — not broadcast to all connected
GPS throttle	Washer emits location every 5s, not every GPS tick	Reduces socket events 10x · saves mobile battery · reduces server load
Redis adapter (Tier 2)	socket.io-redis-adapter via Upstash · all instances share Redis	Horizontal scaling of Socket.io without any app code changes
Sticky sessions	Load balancer routes by socket ID (Railway / AWS ALB)	Long-poll fallback always hits same server · no missed events
Washer location cache	Redis key per washer (TTL 10s) · set by Socket.io · new ID requires per-GPS ping	High DB write per location update bypass Postgres entirely

Caching Strategy

Data	Store	TTL	Invalidation trigger
City list	Redis	24h	Admin updates city table
Company listings	Redis per city+category	5min	Company profile update or new rating
Washer location	Redis per washer	10s	Socket.io GPS event overwrites
Package catalog	TanStack Query	10min	Company edits packages
Customer orders	TanStack Query	30s	Socket.io status event invalidates
Review aggregates	Redis per company	1h	New review submitted

Database Scalability

Strategy	Tool	When to apply
Connection pooling	Neon PgBouncer (built-in transaction mode)	Day 1 — prevents connection exhaustion even at low scale
Prisma pool limit	datasource pool { max: 10 }	Day 1 — caps connections per API instance
Read replica	Neon Pro read replica	Tier 2 — analytics + history queries hit replica, writes hit primary
DB branching	Neon branch per PR	Day 1 — staging database mirrors production schema safely
Gulf region	Neon Bahrain (ap-south-1)	Day 1 — ~30ms latency to UAE/KSA customers
Multi-region (future)	Aurora Global DB primary Bahrain + replica Cairo	Tier 3 — UAE+KSA and Egypt served from closest region

7 BILINGUAL SYSTEM (AR / EN)

Frontend Implementation

Area	Solution	Details
i18n library	next-intl (Next.js) + i18next (Vite/Expo)	Single translation string format across all apps · nested namespaces
URL routing	next-intl middleware	/en/... and /ar/... routes · auto-redirect based on Accept-Language header
RTL layout	dir='rtl' on <html> + Tailwind logical properties	margin-inline-start / padding-inline-end instead of margin-left — works both directions
Arabic font	Cairo (Google Fonts)	Beautiful Arabic typeface · good weight range · loads with next/font for zero CLS
Number format	Intl.NumberFormat per locale	Eastern Arabic numerals (٠١٢٣٤٥٦) in AR mode · AED currency symbol correct per locale
Date format	date-fns with locale param	Arabic Hijri/Gregorian toggle if needed · weekday names in Arabic
Language detection	Device locale → cookie → default EN	First open: device locale checked · user choice persisted to profile + cookie
RTL maps	Mapbox language plugin	Map labels switch to Arabic when AR mode active · Mapbox has full Arabic tile support
RTL animations	CSS logical properties	Transform: translateX mirrors correctly in RTL — no separate RTL animation code

Database Bilingual Fields Convention

Every user-facing content field has two columns — never a JSON blob, always queryable separate columns.

Table	Fields (pattern applied)	Notes
companies	name_en, name_ar, description_en, description_ar	Both required at onboarding · admin reviews both before verification
service_categories	name_en, name_ar	Platform-managed · seeded · never user-generated
cities	name_en, name_ar	Platform-managed · seeded with all UAE + KSA cities
packages	name_en, name_ar, description_en, description_ar	Company enters both during package setup · required fields
addons	name_en, name_ar	Company enters both · short labels only
notifications	Sent in customer.preferred_lang	Push, SMS, and email all respect the user's language preference
reviews	comment stored as-is (no forced translation)	No auto-translation · customer writes in their own language

i18n Translation File Structure

```
packages/i18n/  
  en/  
    common.json ← buttons, labels, errors shared across apps  
    customer.json ← booking flow, tracking, order history  
    company.json ← dashboard labels, analytics, settings  
    washer.json ← job alerts, checklist, earnings  
    notifications.json ← push + SMS + email templates  
  ar/  
    common.json ← same structure, Arabic values  
    customer.json  
    company.json  
    washer.json
```

notifications.json

8 COMPLETE ROADMAP

0

Foundation

Monorepo, infra, CI/CD, design system

Weeks 1–2 · 1 fullstack + 1 designer

INFRA & DEVOPS	DATABASE & API SCAFFOLD	DESIGN SYSTEM
■ Turborepo monorepo scaffold ■ GitHub repo + branch strategy (main/staging/dev) ■ Vercel projects × 4 (customer, company, admin, washer-web) ■ Railway project (API + BullMQ worker) ■ Neon DB — dev + staging + prod branches ■ Upstash Redis — dev + prod ■ GitHub Actions CI pipeline (lint + typecheck + test) ■ Sentry setup on all 5 surfaces ■ Cloudflare R2 bucket setup	■ Prisma schema (all 15 tables) ■ Initial migration + seed data (cities, categories) ■ Fastify server + plugins (cors, helmet, jwt) ■ Zod schemas for all request/response shapes ■ JWT auth middleware (all 4 roles) ■ Role guards (customer / company / washer / admin) ■ Shared TypeScript types package ■ i18n strings package (AR + EN skeleton) ■ API health endpoint + Sentry test	■ Brand identity (logo, color tokens, typography) ■ Figma component library (web + mobile) ■ 8 key screen designs in AR + EN ■ Tailwind config + CSS variables ■ Cairo font integration (Arabic) ■ RTL layout test harness ■ Shared UI components package (buttons, inputs, cards) ■ Storybook for component preview

Deliverables: Working monorepo · CI/CD green · DB migrated · All env vars configured · Design system in Figma

1

MVP — Core Booking Flow

Customer books · company manages · washer executes · real money flows

Weeks 3–8 · 2 fullstack + 1 mobile + 1 designer

CUSTOMER WEB + MOBILE	COMPANY DASHBOARD (WEB)	WASHER APP + API + NOTIFICATIONS
■ Language selection screen (AR/EN first launch) ■ Phone OTP auth — Twilio Verify ■ Home: service category picker (car/carpet/sofa) ■ GPS city detection + manual city fallback ■ Company listing filtered by city + category ■ Company profile: packages + add-ons + reviews ■ Location pin drop with parking note (Mapbox) ■ Quantity selector per service type ■ Checkout with order summary ■ Stripe payment (card + Apple Pay + Google Pay) ■ Order confirmation screen ■ Live tracking screen (Socket.io washer dot on map) ■ Order status progress bar (5 states) ■ Rating + review after completion ■ Tip screen post-completion ■ Order history screen ■ Push notification handling (Expo)	■ Company onboarding: profile setup (EN + AR fields) ■ City coverage multi-select ■ Service category selection ■ Package editor (name EN + AR, price, duration, unit) ■ Add-on editor per category ■ Staff (washer) account management ■ Live order feed (Socket.io real-time) ■ Assign washer to new order ■ Order status management interface ■ Toggle accepting orders on/off ■ Company profile photo upload ■ Operating hours setup ■ Stripe Connect onboarding flow	■ Washer app: phone OTP + PIN setup ■ Go online / offline toggle ■ New job alert screen (accept / decline + 30s timer) ■ Job details view (customer location + parking note) ■ GPS background tracking → Socket.io broadcast ■ Navigation launch (Google Maps deep link) ■ Arrived tap → status update ■ Before photo capture + R2 upload ■ Per-category service checklist ■ Mark complete + after photo upload ■ FCM push (all roles: customer + company + washer) ■ Twilio SMS — order confirmation + arrival alert ■ Resend email — receipt + welcome (AR + EN templates) ■ BullMQ worker — async email + payout jobs ■ Stripe webhook handler (payment capture + refunds)

Deliverables: End-to-end booking works with real money · GPS tracking live · Arabic + English fully functional · Ready for private beta

MILESTONE: PRIVATE BETA LAUNCH

2

Growth Features

Loyalty, promos, analytics, admin panel, WhatsApp

Weeks 9–14 · 2 fullstack + 1 mobile

CUSTOMER RETENTION ■ Loyalty points wallet (earn per order) ■ Points redemption at checkout ■ Promo codes + discount engine (% and fixed) ■ Referral program (customer gets credit) ■ In-app wallet credit top-up (Stripe) ■ 1-tap rebook from order history ■ Scheduled bookings (future date + time slot) ■ In-app chat (customer ↔ company) ■ WhatsApp confirmation (360dialog) ■ Push notification preferences settings	COMPANY TOOLS ■ Revenue analytics dashboard ■ Orders by day / week / month chart ■ Peak hours heatmap ■ Popular packages + add-ons report ■ Customer repeat rate metric ■ Washer performance scores ■ Payout history + PDF invoice download ■ Company-level promo code creation ■ Holiday / unavailability schedule ■ Multi-city toggle (enable/disable per city)	ADMIN PANEL (FULL) ■ Company onboarding review + verify workflow ■ Platform-wide analytics (GMV, orders, cities) ■ City + category management (add new cities) ■ Commission rate editor per company ■ Dispute management with photo evidence viewer ■ Featured company placement + paid slots ■ Platform promo campaign creator ■ Review moderation queue ■ Washer fraud flag system ■ Manual refund tool ■ System health dashboard ■ Audit log viewer (all admin actions)
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Deliverables: Public launch ready · Loyalty + promos live · Admin has full control · Company self-service · WhatsApp notifications

MILESTONE: PUBLIC LAUNCH

3

Scale & Intelligence

AI features, subscriptions, B2B, new markets, infra hardening

Weeks 15–24 · 3 fullstack + 1 mobile + 1 data

AI + SMART FEATURES (Claude API) ■ Smart price recommendations per city/season ■ Review sentiment analysis (flag negative trends) ■ Personalized company ranking per user ■ Demand prediction for peak slot pricing ■ Automatic before/after photo quality check ■ AI customer support chatbot (AR + EN) ■ Anomaly detection (GPS fraud, fake reviews) ■ Smart notification timing (open-rate optimization)	SUBSCRIPTIONS + B2B ■ Monthly subscription plans for customers ■ 4 washes/mo plan, 8 washes/mo plan ■ Fleet accounts (companies with car pools) ■ Corporate billing (monthly invoice) ■ Multi-vehicle profiles per account ■ Subscription analytics dashboard ■ B2B sales admin tools ■ SLA guarantees for corporate accounts	PLATFORM + EXPANSION ■ Expand to KSA cities (Riyadh, Jeddah, Dammam) ■ Expand to Egypt (Cairo, Alexandria) ■ Neon read replica activation ■ Socket.io Redis adapter (multi-instance) ■ Load testing with k6 before each expansion ■ App Store + Google Play listing optimization ■ Performance budget monitoring (Core Web Vitals) ■ Multi-region readiness assessment ■ Mattress + curtain cleaning (new categories) ■ API public docs for enterprise integrations
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Deliverables: AI-powered platform · Subscription revenue stream · Multi-country · Infrastructure ready for 5,000+ orders/day

MILESTONE: REGIONAL SCALE — KSA + EGYPT

9 TEAM, COST & REVENUE MODEL

Recommended Team Composition

Role	Phase	Responsibilities
Fullstack Developer (Lead)	All phases	API + DB + company dashboard + admin panel · owns architecture decisions
Fullstack Developer	Phase 1+	Customer web (Next.js) · bilingual implementation · Stripe integration
React Native Developer	Phase 1+	Customer mobile + washer mobile · Expo · GPS + camera + push
UI/UX Designer	Phase 0–2	Figma design system · all screen designs AR + EN · handoff to devs
QA Engineer (part-time)	Phase 1+	Test cases · regression · bilingual UI testing · device matrix
Product Owner / You	All phases	Requirements · company onboarding · market relationships · prioritization
Data Engineer (Phase 3)	Phase 3	AI features · analytics pipeline · subscription metrics

Infrastructure Cost Estimate

Service	Tier 1 (Launch)	Tier 2 (Growth)	Tier 3 (Scale)
Vercel (4 frontends)	\$0 (hobby)	\$80/mo (Pro)	\$150/mo
Railway (API + worker)	\$20/mo	\$80/mo	\$200/mo (or migrate to AWS)
Neon PostgreSQL	\$0 (free)	\$69/mo (Pro)	\$200/mo+ (Scale)
Upstash Redis	\$0 (free)	\$30/mo	\$100/mo
Cloudflare R2	~\$2/mo	~\$15/mo	~\$50/mo
Twilio (SMS + Verify)	~\$10/mo	~\$80/mo	~\$300/mo
Resend (email)	\$0 (free)	\$20/mo	\$80/mo
Stripe fees	2.9% + 30¢/txn	2.9% + 30¢/txn	Negotiated rate
Expo EAS Build	\$0	\$29/mo	\$99/mo
Sentry	\$0 (free)	\$26/mo	\$80/mo
TOTAL (approx)	~\$50/mo	~\$350/mo	~\$2,000+/mo

Revenue Model

Stream	Rate	Notes
Platform commission	15–20% per order	Primary revenue · auto-deducted before company payout via Stripe Connect
Featured listing	AED 500/company/month	Top placement in city+category search results
Customer subscription	AED 99/month	4 washes included + discount on extras · reduces per-order Stripe fees
Corporate billing	Custom / AED 1,000+/mo	Fleet accounts, hotel groups, car rental companies
Promo placements	AED 200/campaign	Company-sponsored banners on home screen
New city onboarding	AED 1,000 one-time	Setup fee for company to activate a new city coverage

Timeline Summary

Phase	Duration	Key Milestone	Cumulative
Phase 0 — Foundation	2 weeks	Monorepo + CI/CD live	Week 2
Phase 1 — MVP	6 weeks	Private beta (real orders)	Week 8
Phase 2 — Growth	6 weeks	Public launch	Week 14
Phase 3 — Scale	10 weeks	KSA + Egypt expansion	Week 24
Total to public launch	~14 weeks	—	~3.5 months
Total to regional scale	~24 weeks	—	~6 months

10 QUICK START CHECKLIST — WEEK 1 ACTIONS

Do these first — before writing a single line of product code.

Accounts to create (Day 1)

- GitHub organization + monorepo
- Vercel account + 4 projects (customer-web, company-web, admin-web, washer-web)
- Railway account + project
- Neon account + project (create dev, staging, prod branches)
- Upstash account + Redis (dev + prod databases)
- Cloudflare account + R2 bucket (cleanly-photos)
- Stripe account + enable Connect + get API keys
- Twilio account + buy UAE/AED virtual number + enable Verify
- Resend account + verify sending domain
- Sentry account + 5 projects (one per app surface)
- Mapbox account + access token
- Google Cloud project + Maps SDK enabled
- Expo account + EAS CLI installed
- 360dialog account (WhatsApp — takes 2–3 weeks to verify, start now)

Code to scaffold (Week 1)

- `npx create-turbo@latest cleanly-monorepo`
- Add apps/api — Fastify + Prisma setup
- Add apps/customer-web — Next.js 14 + next-intl + Tailwind
- Add apps/company-web — Vite + React + TanStack Query
- Add apps/customer-mobile — `npx create-expo-app`
- Add apps/washer-mobile — Expo (can reuse customer-mobile base)
- Add apps/admin-web — Next.js + NextAuth
- Add packages/shared-types — TypeScript interfaces
- Add packages/i18n — translation strings (en/ + ar/)
- Add packages/ui-components — shared Button, Card, Input
- Add prisma/schema.prisma — all 15 tables
- Run first migration on Neon dev branch
- Set up GitHub Actions — CI runs on every PR

Critical path items (Week 2)

- Get Twilio Verify working end-to-end (OTP send + verify)
- Get Stripe payment working in test mode
- Get Socket.io connecting between API and a test client
- Get Expo app running on a real device (not simulator)
- Verify Mapbox renders Arabic labels correctly
- Verify RTL layout flips correctly on customer-web with next-intl
- Get first Neon DB query working through Prisma
- Set up Sentry and confirm errors are being captured

- Add seed data: UAE cities, 3 service categories, 1 test company
- First end-to-end test: create order in DB, read from API, display in web

Submit immediately (takes weeks to process)

- TikTok Content API app review (if social needed) — 2-4 weeks
- Apple Developer account + App Store Connect setup — can take days
- Google Play Console account setup — usually fast
- 360dialog WhatsApp Business verification — 2-3 weeks
- UAE trade license verification for Stripe (some accounts need this)
- App name trademark search — do before naming anything publicly